
SESSION-1

Task1:- Installation of WSL and GCC compiler

As i am running these commands using UBUNTU OS

```
sujit@sujit-LOQ-15IRX9:~$ getent passwd "username" | cut -d ':' -f 5 | cut -d ',' -f 1
sujit@sujit-LOQ-15IRX9:~$ getent passwd $USER | cut -d ':' -f 5 | cut -d ',' -f 1
sujit
sujit@sujit-LOQ-15IRX9:~$
```

GCC installation

```
sujit@sujit-LOQ-15IRX9:~$ sudo apt update
sujit@sujit-LOQ-15IRX9:~$ sudo apt install --reinstall gcc g++ build-essential
[sudo: authenticated] Password:
Get:1 https://brave-browser-apt-release.s3.brave.com stable InRelease [7,547 B]
Get:2 https://dl.google.com/linux/chrome-stable/deb stable InRelease [2,548 B]
Get:3 https://brave-browser-apt-release.s3.brave.com stable/main amd64 Packages [35.9 kB]
Get:4 https://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Get:5 https://brave-browser-apt-release.s3.brave.com stable/main arm64 Packages [35.8 kB]
Get:6 https://dl.google.com/linux/chrome-stable/deb stable/main amd64 Packages [1,421 B]
Hit:7 http://in.archive.ubuntu.com/ubuntu resolute InRelease
Get:8 http://in.archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:9 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [394 kB]
Get:10 http://security.ubuntu.com/ubuntu resolute-security/main Translation-en [94.3 kB]
Get:11 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [46.7 kB]
Get:12 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Packages [158 kB]
Get:13 http://security.ubuntu.com/ubuntu resolute-security/universe Translation-en [51.1 kB]
Get:14 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [43.1 kB]
Hit:15 http://in.archive.ubuntu.com/ubuntu resolute-backports InRelease
Get:16 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [485 kB]
Get:17 http://in.archive.ubuntu.com/ubuntu resolute-updates/main Translation-en [117 kB]
Get:18 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [97.8 kB]
Get:19 http://in.archive.ubuntu.com/ubuntu resolute-updates/restricted amd64 Packages [359 kB]
Get:20 http://in.archive.ubuntu.com/ubuntu resolute-updates/restricted Translation-en [69.7 kB]
Get:21 http://in.archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [238 kB]
Get:22 http://in.archive.ubuntu.com/ubuntu resolute-updates/universe Translation-en [70.3 kB]
Get:23 http://in.archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Components [182 kB]
Fetched 2,773 kB in 13s (209 kB/s)
154 packages can be upgraded. Run 'apt list --upgradable' to see them.
The following packages were automatically installed and are no longer required:
  linux-headers-7.0.0-14 linux-headers-7.0.0-14-generic linux-image-unsigned-7.0.0-14-generic linux-kernel-modules-7.0.0-14-generic linux-modules-7.0.0-14-generic linux-tools-7.0.0-14-generic
Use 'sudo apt autoremove' to remove them.

Installing:
  build-essential g++

Installing dependencies:
  binutils fakeroot g++-15-x86-64-linux-gnu libalgorithm-diff-perl libalgorithm-merge-perl libdpkg-perl libfile-fcntllock-perl lto-disabled-list
  dpkg-dev g++-15 g++-x86-64-linux-gnu libalgorithm-diff-xs-perl libcrypt-dev libfakeroot libstdc++-15-dev make

Suggested packages:
  bzr2-doc debconf debconf-utils debconf-keyring g++-multilib g++-15-multilib gcc-15-doc bzr libstdc++-15-doc make-doc

Summary:
  Upgrading: 0, Installing: 18, Reinstalling: 1, Removing: 0, Not Upgrading: 154
  Download size: 19.3 MB
  Space needed: 73.4 MB / 166 GB available

Continue? [Y/n] Y
Get:1 http://in.archive.ubuntu.com/ubuntu resolute/main amd64 libstdc++-15-dev amd64 15.2.0-16ubuntu1 [2,920 kB]
dx [1 libstdc++-15-dev 1,955 kB/2,920 kB 67%] 100 kB/s 1min 33s
```

```

Unpacking libltdl-dev:amd64 (2.4.4-13) ...
Selecting previously unselected package dpkg-dev.
Preparing to unpack ./dpkg-dev_1.23.7ubuntu1_all.deb.
Unpacking dpkg-dev (1.23.7ubuntu1) ...
Selecting previously unselected package libcrypt-dev:amd64.
Preparing to unpack ./libcrypt-dev_1:4.4.0-1ubuntu1_amd64.deb.
Unpacking libcrypt-dev:amd64 (1:4.4.0-1ubuntu1) ...
Selecting previously unselected package build-essential.
Preparing to unpack ./build-essential_12.1ubuntu2.26.04.2_amd64.deb.
Unpacking build-essential (12.1ubuntu2.26.04.2) ...
Selecting previously unselected package libfakeroot:amd64.
Preparing to unpack ./libfakeroot_1.37.2-1_amd64.deb.
Unpacking libfakeroot:amd64 (1.37.2-1) ...
Selecting previously unselected package fakeroot.
Preparing to unpack ./fakeroot_1.37.2-1_amd64.deb.
Unpacking fakeroot (1.37.2-1) ...
Preparing to unpack ./gcc-4.8-ubuntu1.2.0-Subunit1_amd64.deb.
Unpacking gcc (4:15.2.0-Subunit1) over (4:15.2.0-Subunit1) ...
Selecting previously unselected package libalgorithm-diff-perl.
Preparing to unpack ./libalgorithm-diff-perl_1.201-1_all.deb.
Unpacking libalgorithm-diff-perl (1.201-1) ...
Selecting previously unselected package libalgorithm-diff-xs-perl.
Preparing to unpack ./libalgorithm-diff-xs-perl_0.04-9_amd64.deb.
Unpacking libalgorithm-diff-xs-perl (0.04-9) ...
Selecting previously unselected package libalgorithm-merge-perl.
Preparing to unpack ./libalgorithm-merge-perl_0.08-5_all.deb.
Unpacking libalgorithm-merge-perl (0.08-5) ...
Selecting previously unselected package libfile-fcntllock-perl.
Preparing to unpack ./libfile-fcntllock-perl_0.22-4ubuntu2_amd64.deb.
Unpacking libfile-fcntllock-perl (0.22-4ubuntu2) ...
Setting up lto-disabled-list (79) ...
Setting up libfile-fcntllock-perl (0.22-4ubuntu2) ...
Setting up libalgorithm-diff-perl (1.201-1) ...
Setting up bzip2 (1.0.8-4build2) ...
Setting up libfakeroot:amd64 (1.37.2-1) ...
Setting up gcc (4:15.2.0-Subunit1) ...
Setting up fakeroot (1.37.2-1) ...
update-alternatives: using /usr/bin/fakeroot-sysv to provide /usr/bin/fakeroot (fakeroot) in auto mode
Setting up libstdc++-15-dev:amd64 (15.2.0-10ubuntu1) ...
Setting up make (4.4.1-3) ...
Setting up libdpkg-perl (1.23.7ubuntu1) ...
Setting up libcrypt-dev:amd64 (1:4.4.0-1ubuntu1) ...
Setting up libalgorithm-diff-xs-perl (0.04-9) ...
Setting up libalgorithm-merge-perl (0.08-5) ...
Setting up g++-15-x86_64-linux-gnu (15.2.0-10ubuntu1) ...
Setting up dpkg-dev (1.23.7ubuntu1) ...
Setting up g++-x86_64-linux-gnu (4:15.2.0-Subunit1) ...
Setting up g++-15 (15.2.0-Subunit1) ...
Setting up g++ (4:15.2.0-Subunit1) ...
update-alternatives: using /usr/bin/g++ to provide /usr/bin/c++ (c++) in auto mode
Setting up build-essential (12.1ubuntu2.26.04.2) ...
Processing triggers for man-db (2.13.1-1build3) ...
Processing triggers for libc-bin (2.43-2ubuntu2.3) ...
root@kali:~# dpkg-query -f='${Package} ${Version} ${Architecture}\n' -W -f='${Package} ${Version} ${Architecture}\n' | grep fakeroot
fakeroot 1.37.2-1 amd64

```

Task 2: To Analyse process abstraction, execute `fork()`, understand `exec()` family, analyse parent-child relationships, inspect process tree, practice system call tracing.

CODE:

```
GNU nano 8.7.1
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <stdlib.h>

int main() {
    pid_t pid;

    printf("Parent Process Started\n");
    printf("Parent PID: %d\n", getpid());

    pid = fork();

    if (pid < 0) {
        perror("fork failed");
        return 1;
    }

    if (pid == 0) {
        printf("\nChild Process\n");
        printf("Child PID: %d\n", getpid());
        printf("Parent PID: %d\n", getppid());
        printf("Executing ls using exec()\n");
        execlp("ls", "ls", "-l", NULL);
        perror("exec failed");
        exit(1);
    } else {
        printf("\nParent Process\n");
        printf("Parent PID: %d\n", getpid());
        printf("Child PID: %d\n", pid);
        printf("Parent waiting for child...\n");
        wait(NULL);
        printf("Child process completed.\n");
    }

    return 0;
}
```

OUTPUT:

```
sujit@sujit-LOQ-15IRX9: ~  
  
sujit@sujit-LOQ-15IRX9:~/Documents/GitHub/OSSP_2520090085/Skills$ nano main4.c  
sujit@sujit-LOQ-15IRX9:~/Documents/GitHub/OSSP_2520090085/Skills$ gcc main4.c -o main4  
sujit@sujit-LOQ-15IRX9:~/Documents/GitHub/OSSP_2520090085/Skills$ ./main4  
Parent Process Started  
Parent PID: 32841  
  
Parent Process  
Parent PID: 32841  
Child PID: 32842  
Parent waiting for child...  
  
Child Process  
Child PID: 32842  
Parent PID: 32841  
Executing ls using exec()  
total 2220  
-rw-rw-r-- 1 sujit sujit 717983 Aug 20 14:21 'OSSP SWeek-4.docx.pdf'  
-rw-rw-r-- 1 sujit sujit 745432 Aug 20 14:53 'OSSP Skill Week2 .docx.pdf'  
-rwxrwxr-x 1 sujit sujit 16288 Aug 20 15:27 main4  
-rw-rw-r-- 1 sujit sujit 870 Aug 20 15:27 main4.c  
-rw-rw-r-- 1 sujit sujit 783873 Aug 20 15:14 'ossP SWeek-3.docx.pdf'  
Child process completed.  
sujit@sujit-LOQ-15IRX9:~/Documents/GitHub/OSSP_2520090085/Skills$
```

SESSION-2

Task 1: To Create Main Loop, Display Prompt, Read User Input, Handle Exit Conditions, Design Control Flow Diagram, Test Interactive Loop

A) Create a file by using

```
sujit@sujit-LOQ-15IRX9: ~  
  
sujit@sujit-LOQ-15IRX9:~/Documents/GitHub/OSSP_2520090085/Skills$ nano main5.c  
sujit@sujit-LOQ-15IRX9:~/Documents/GitHub/OSSP_2520090085/Skills$
```

Then press enter and the code terminal will open then write a code

```
GNU nano 8.7.1
#include <stdio.h>
#include <string.h>

int main() {
    char input[100];

    while (1) {
        printf("myShell> ");
        fgets(input, sizeof(input), stdin);

        // Remove newline
        input[strcspn(input, "\n")] = 0;

        // Exit condition
        if (strcmp(input, "exit") == 0) {
            printf("Exiting Shell...\n");
            break;
        }

        printf("You entered: %s\n", input);
    }

    return 0;
}
```

After writing the code save the your program by ctrl O and for coming back to command press ctrl X then compile it by the command

```
sujit@sujit-LOQ-15IRX9:~/Documents/GitHub/OSSP_2520090085/Skills$ nano main5.c
sujit@sujit-LOQ-15IRX9:~/Documents/GitHub/OSSP_2520090085/Skills$ nano main5.c
sujit@sujit-LOQ-15IRX9:~/Documents/GitHub/OSSP_2520090085/Skills$ gcc main5.c -o main5
```

After compiling to get the output enter command

```
sujit@sujit-L0Q-15IRX9:~/Documents/GitHub/OSSP_2520090085/Skills$ nano main5.c
sujit@sujit-L0Q-15IRX9:~/Documents/GitHub/OSSP_2520090085/Skills$ nano main5.c
sujit@sujit-L0Q-15IRX9:~/Documents/GitHub/OSSP_2520090085/Skills$ gcc main5.c -o main5
sujit@sujit-L0Q-15IRX9:~/Documents/GitHub/OSSP_2520090085/Skills$ ./main5
myShell> OSSP
You entered: OSSP
myShell> POWERSHELL
You entered: POWERSHELL
myShell> UBUNTU
You entered: UBUNTU
myShell> 
```

Task 2: To Capture Keyboard Input, Handle Backspace, Process Enter Key, Manage Input Buffer, Support Multi-Character Commands, Test User Interaction.

CODE:

```

GNU nano 8.7.1
#include <stdio.h>

int main() {
    char buffer[100];
    int ch;
    int index = 0;

    printf("Type a command and press Enter:\n");
    printf("input> ");

    while ((ch = getchar()) != '\n' && ch != EOF) {
        if (ch == 127 || ch == 8) {
            if (index > 0) {
                index--;
            }
        } else if (index < 99) {
            buffer[index++] = ch;
        }
    }

    buffer[index] = '\0';

    printf("\nCommand entered: %s\n", buffer);
    printf("Characters stored: %d\n", index);

    return 0;
}

```

OUTPUT-:

```

sujit@sujit-L0Q-15IRX9:~/Documents/GitHub/OSSP_2520090085/Skills$ nano main7.c
sujit@sujit-L0Q-15IRX9:~/Documents/GitHub/OSSP_2520090085/Skills$ gcc main7.c -o main7
sujit@sujit-L0Q-15IRX9:~/Documents/GitHub/OSSP_2520090085/Skills$ ./main7
Type a command and press Enter:
input> Hello World

Command entered: Hello World
Characters stored: 11
sujit@sujit-L0Q-15IRX9:~/Documents/GitHub/OSSP_2520090085/Skills$ 

```